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# Properties of Materials
# Lecture 8

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|---|
| **Lecture 5**: Step 1 - Translation, Step 2 - Screening, Materials Selection Process (2), Step 3 - Ranking Materials, Example - Hydraulic Cylinder Tie Rods, Step 4 - Refining Selection, Concepts and Terminology (2) |
| **Lecture 6**: The Elastic Moduli, Deformation of Brittle Materials, Deformation of Ductile Materials, Hooke's Law - Young's Modulus, Linear Elastic Behaviour, Structural Engineering, What Might Have Saved the Bridge, Engineering Stress and Strain, True Stress and Strain, Comparison |
| **Lecture 7**: Practice Time, Definitions, Concepts and Terminology (3), Task |
| **Exercise Seminar 1** |
| **Lecture 8**: Bonding Between Atoms, Learning Objectives, Bonding Energies, Primary Bonding, Ionic Bonding, Covalent Bonding, Metallic Bonding, Crystal Structures, Body-Centred Cubic, Face-Centred Cubic, Secondary Bonding, Van der Waals Forces, Hydrogen Bonding, Bonding Energies and Properties, Bonding Types and Properties, Concepts and Terminology (4) |

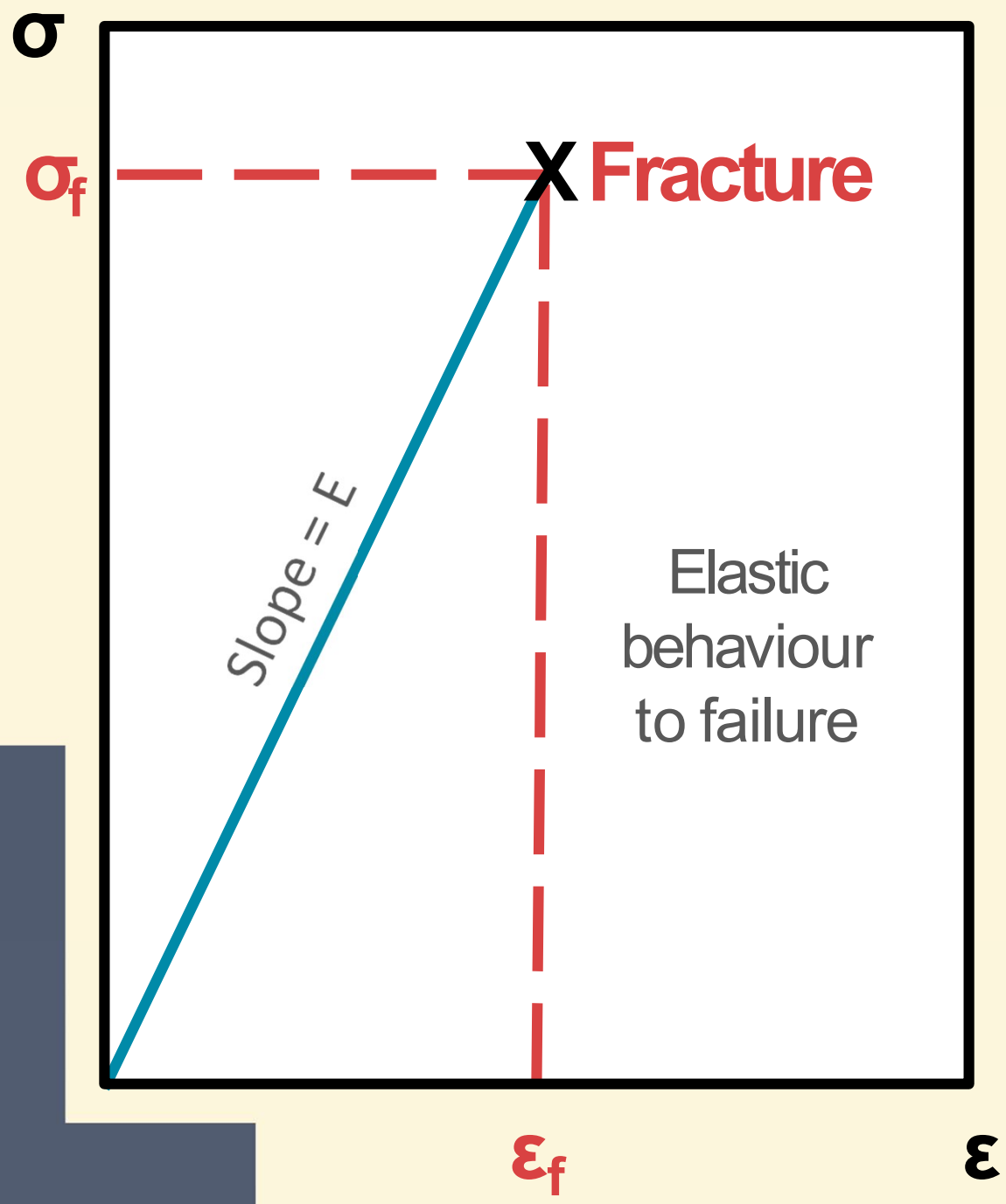
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# Bonding between atoms

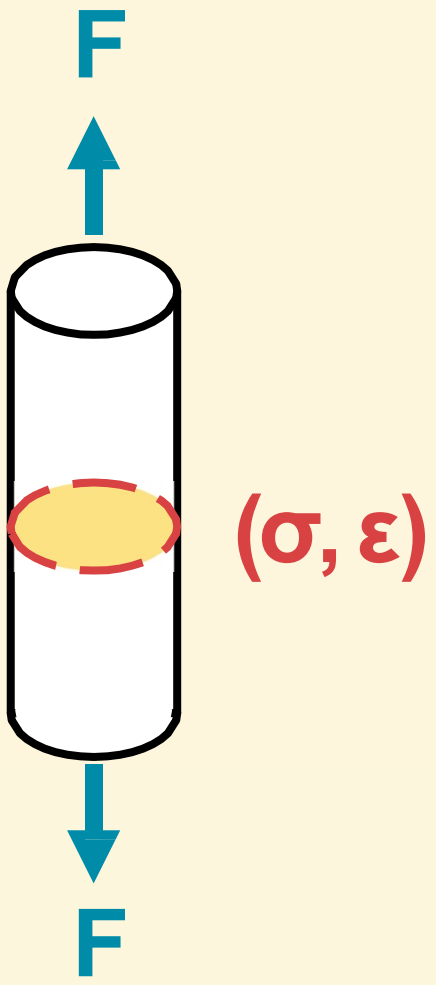
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# Recap from last time...

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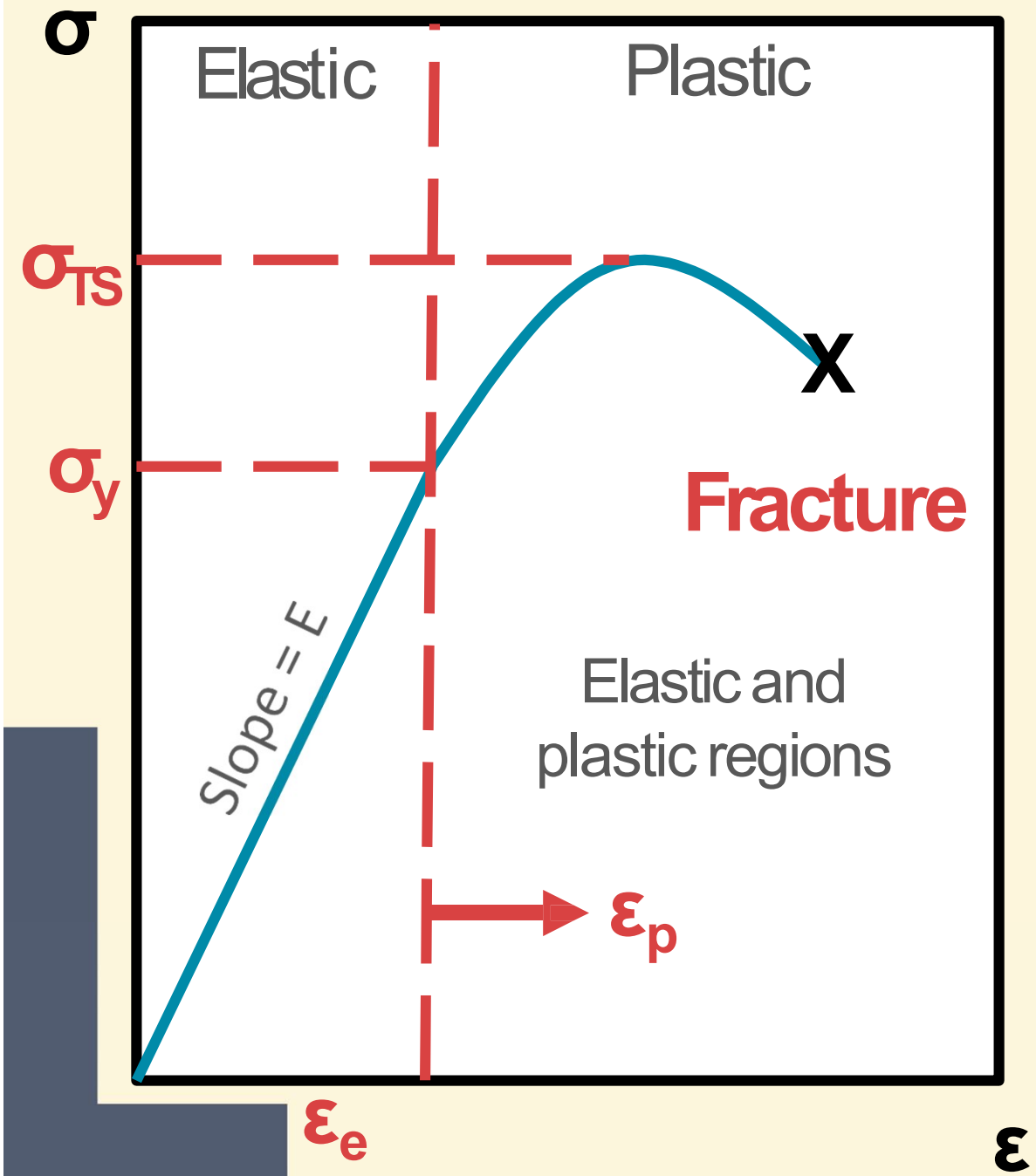



## Deformation of brittle materials

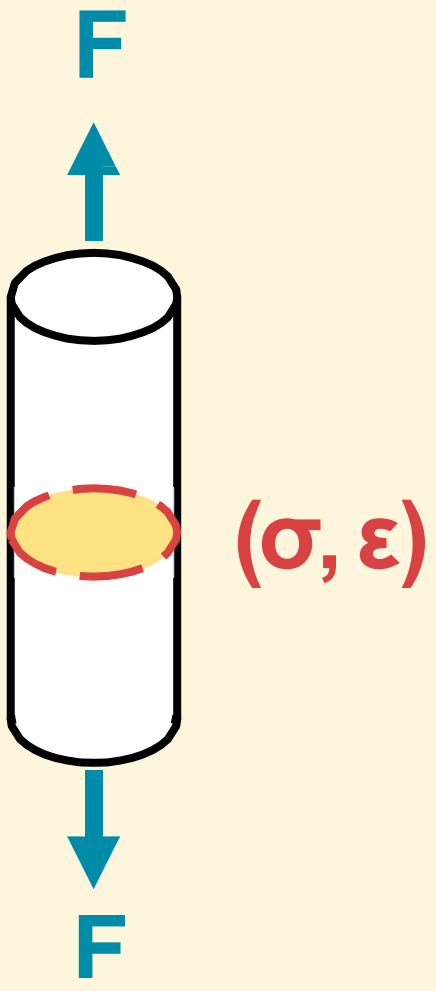


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Total strain, $\varepsilon = \varepsilon_e + \varepsilon_p = \frac{\sigma_y}{E} + \varepsilon_p$




# Deformation of ductile materials



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# Linear elastic behaviour

- Ball and spring model of elasticity:
- Material in tension –bonds stretched and atoms move apart
- Material in compression –bonds compressed and atoms move closer together

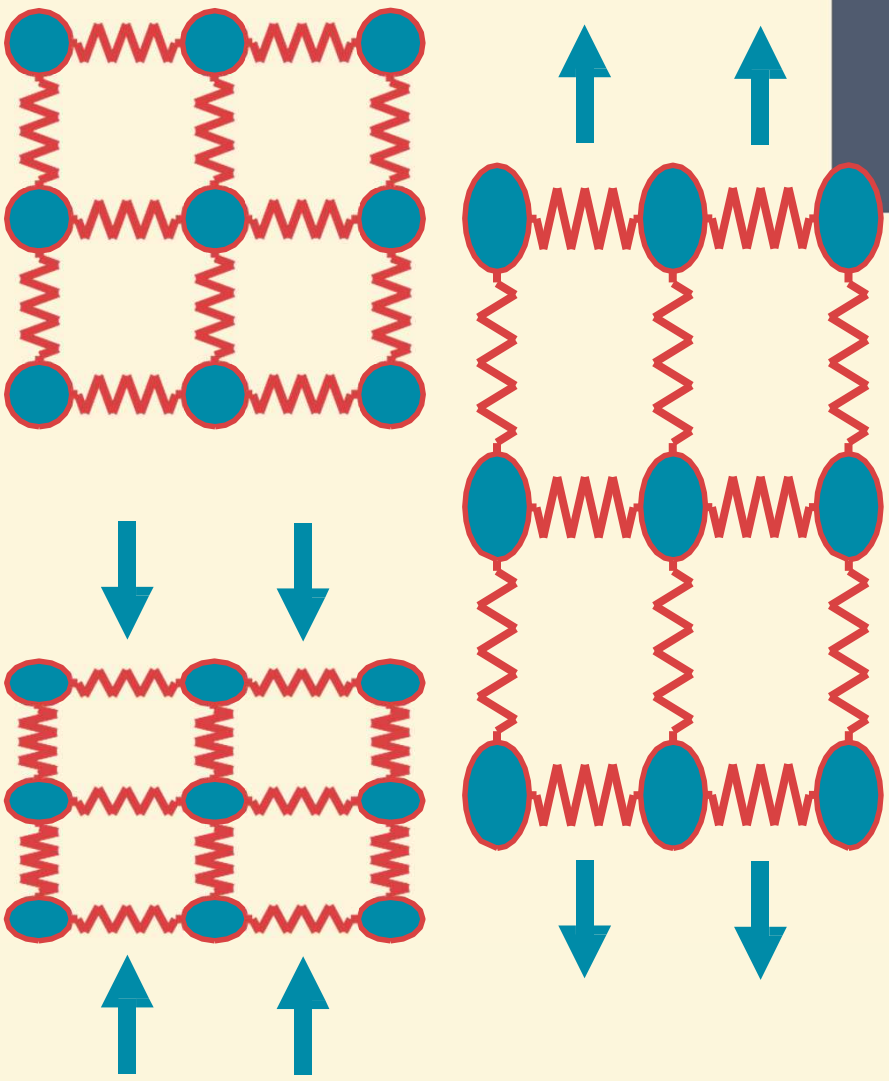


Illustration adapted from: Gordon, J.E., (1978), *Structures or Why Thing's Don't Fall Down*

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# Learning objectives

- Develop an appreciation of the different ways in which atoms can bond together and how that begins to influence material properties

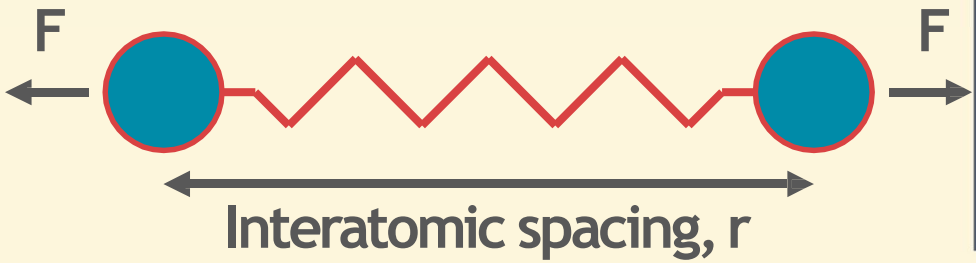
- Expand our ball and spring model to consider **bond energies** and how these vary with bond type

- Learn about **Primary** and **Secondary** bonding mechanisms:
  - Ionic, covalent and metallic
  - Van der Waals forces and hydrogen bonding

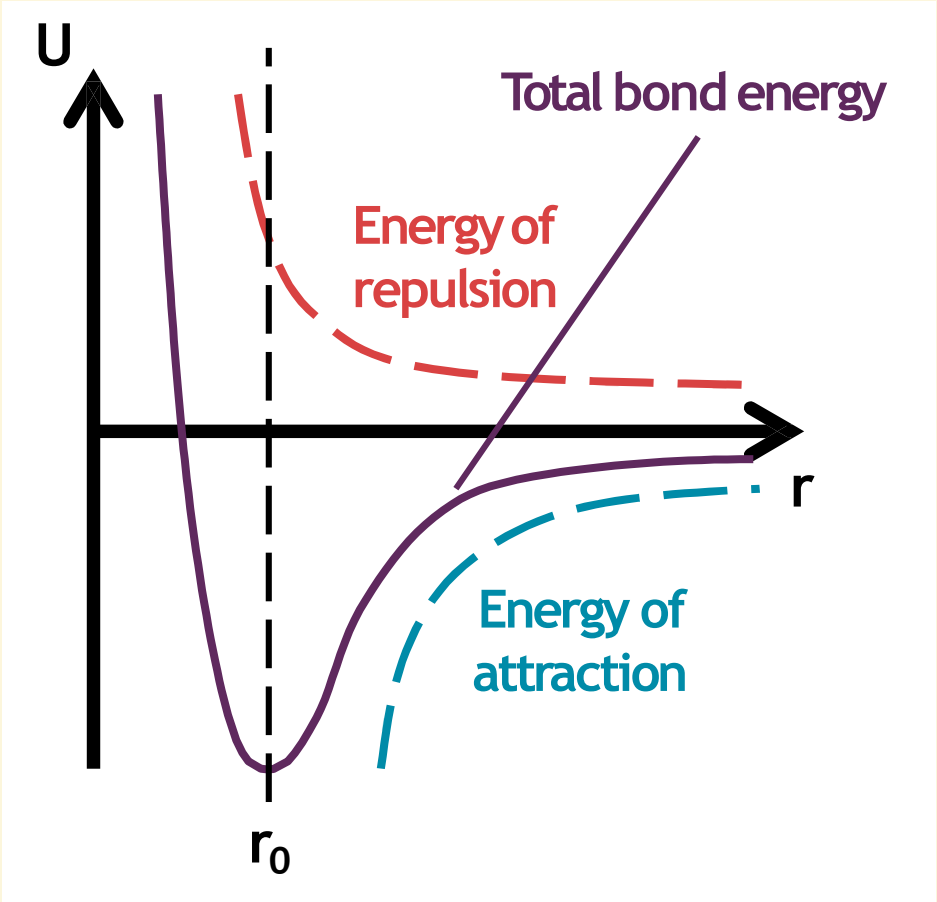
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# Bond energies (1)

- Ball and spring model:
  - Atoms subjected to competing repulsive and attractive forces

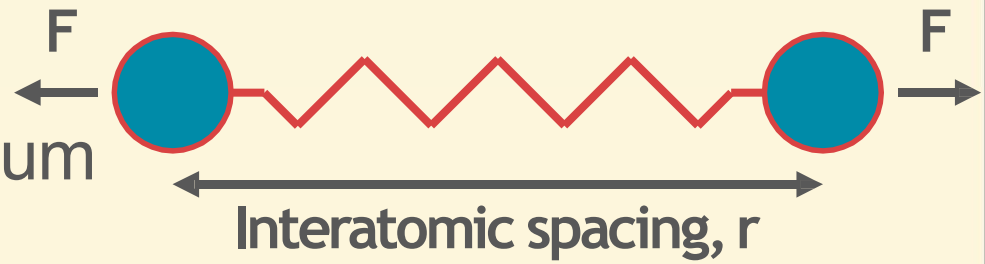



- At rest, atoms will be in their lowest energy state

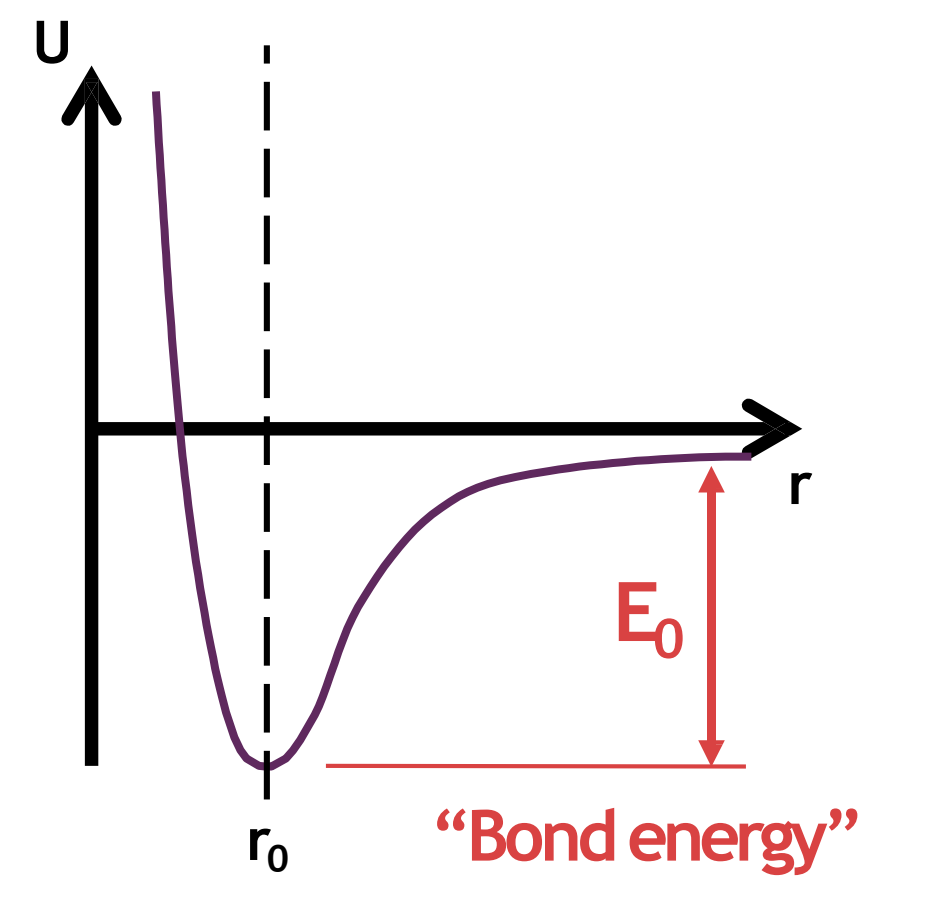



- $r_0$ = equilibrium interatomic spacing
  $U$ = potential energy of bond

# Bond energies (2)




- **Bond energy ($E_0$)** is the minimum of potential vs spacing curve
- Bond energy describes the energy required to break the bond between (dissociate) atoms
- Depth of the 'well' in the total energy curve gives bond strength:
  - Deep well = strong
  - Shallow well = weak



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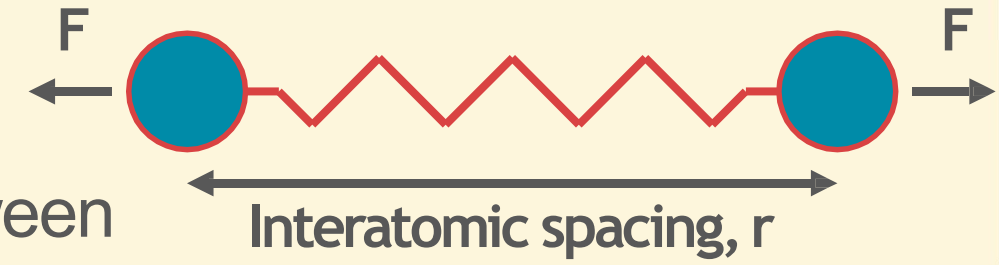
# Primary bonding

- In Chemistry, you may have previously discussed primary bonds as being 'intra-molecular' bonds:
  - In Engineering Materials, we don't deal with many molecular substances!

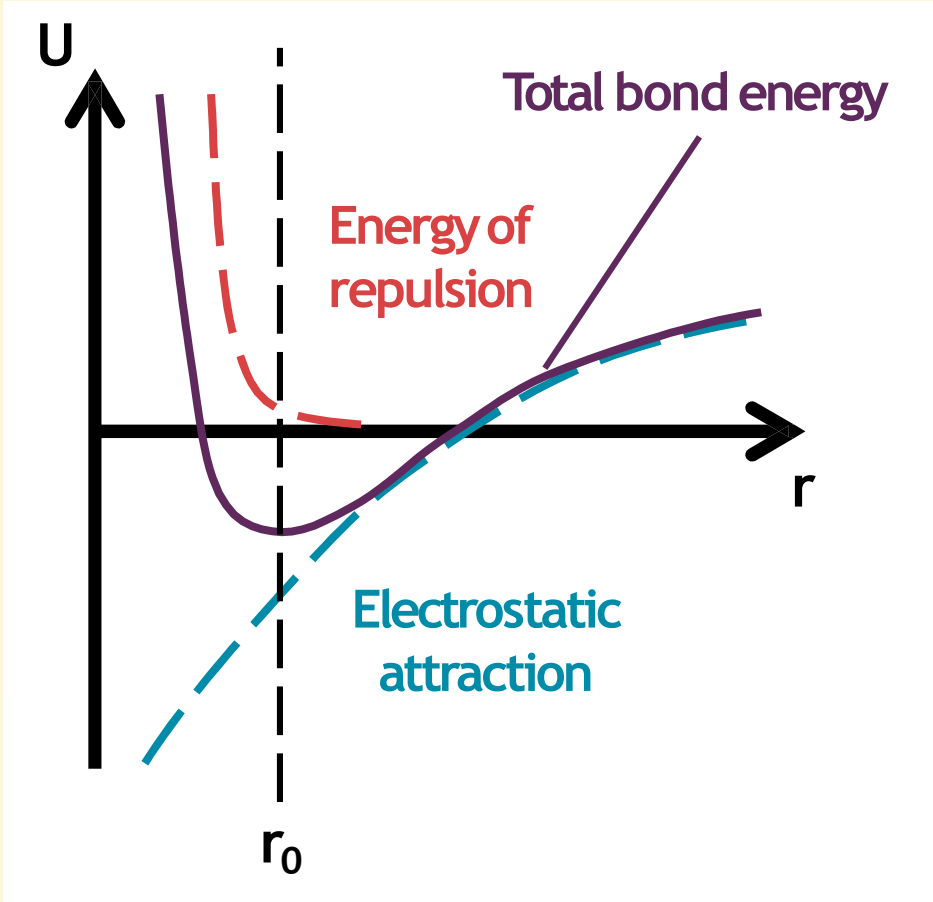
- Primary bond types:
  - **Ionic** bonds – electrostatic attraction between positively and negatively charged ions
  - **Covalent** bonds – sharing of electron orbitals between neighbouring nuclei
  - **Metallic** bonds – high energy electrons leave their parent atoms to create a 'sea' of electrons

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# Ionic bonding (1)




- **Electrostatic attraction** between ions:
  - e. g . $Na^+ + Cl^- \longrightarrow NaCl$
- At equilibrium spacing, electrostatic attraction balanced with **repulsion energy**:
  - Ions repulse when electron charge distributions start to overlap
- Ionic bonds are **non-directional** but ionic structures must alternate from +ve to -ve

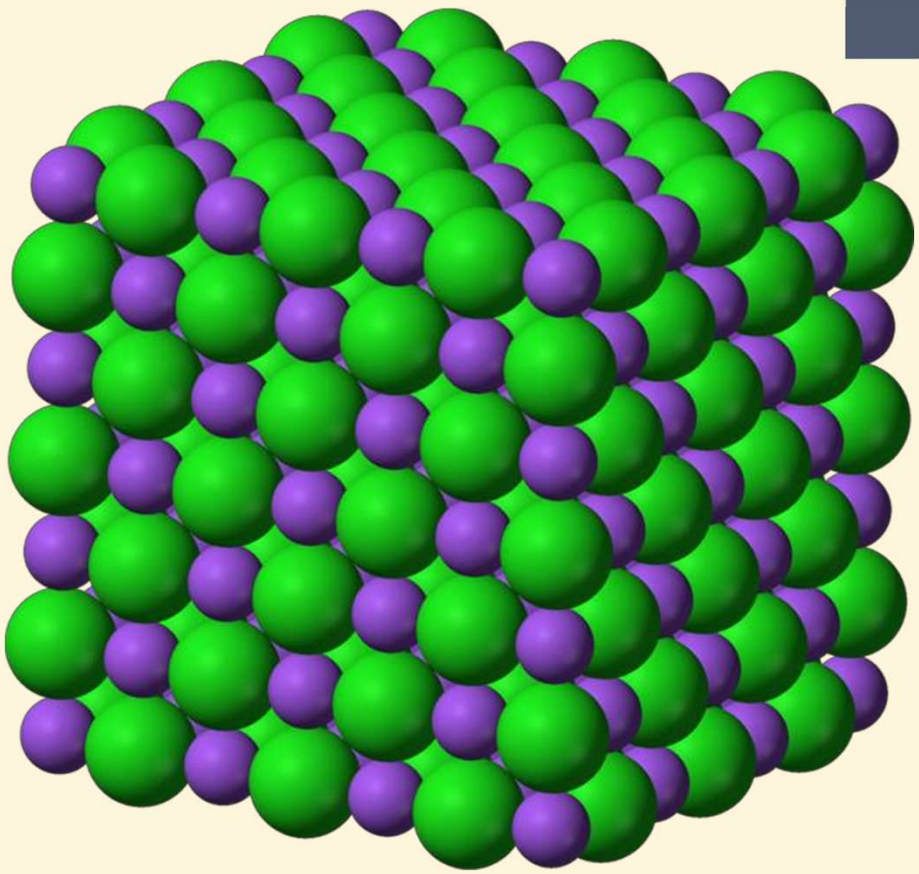


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# Ionic bonding (2)

- Common examples:
  - Table salt (NaCl) and other chlorides
  - Metal oxides (e.g. MgO, $Al_2O_3$)
  - Many ceramics (although these also contain covalent bonding)

- Ionic bonds tend to be **stronger than** covalent or metallic bonds

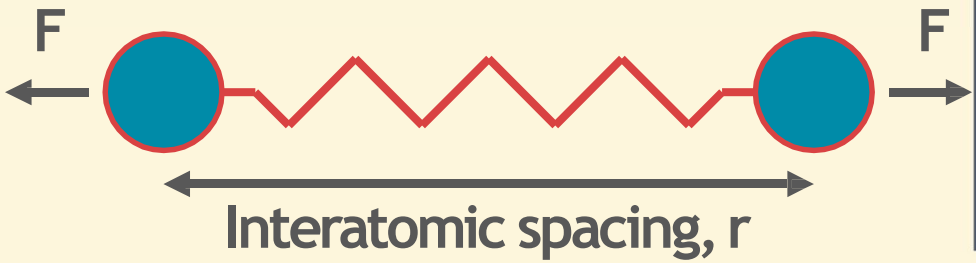


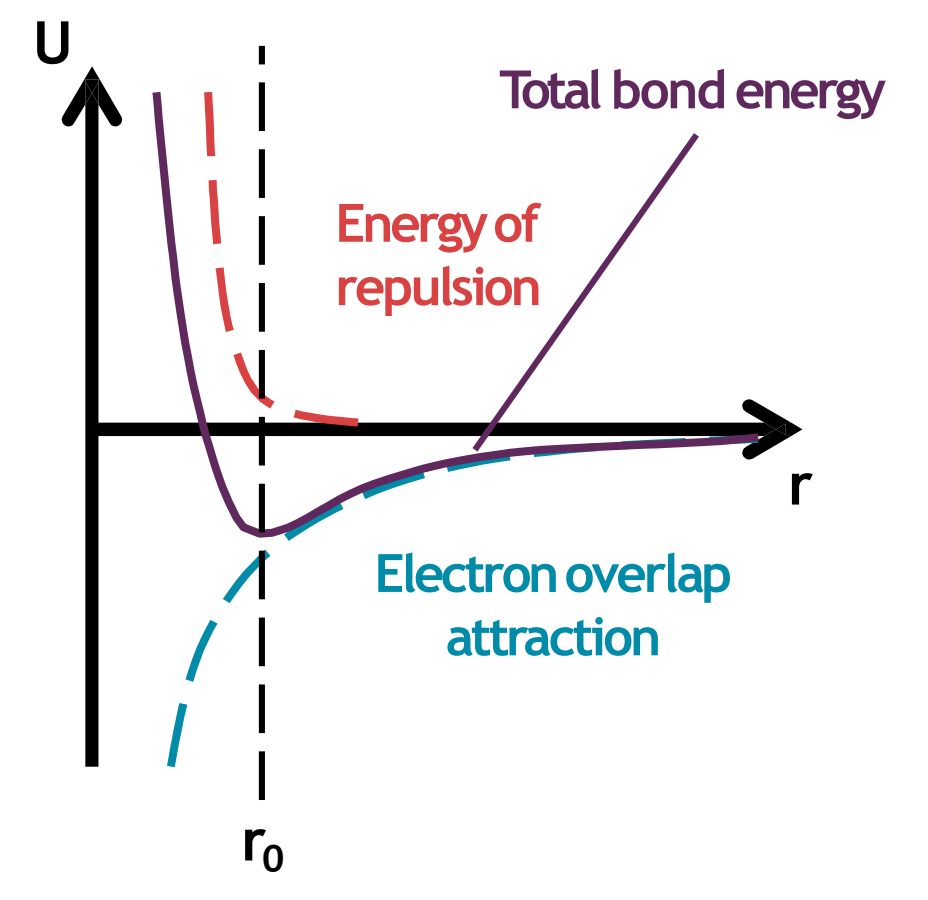
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# Covalent bonding

- **Sharing of electron orbitals**:
  - e.g. Diamond, silicon carbide
  - Contributes to bonding of high-melting point metals (e.g. W, Mo, Ta)
  - Bonding in hydrocarbon chains

- Covalent bonds are **highly directional**:
  - Influences the way in which atoms 'stack' together to form larger structures



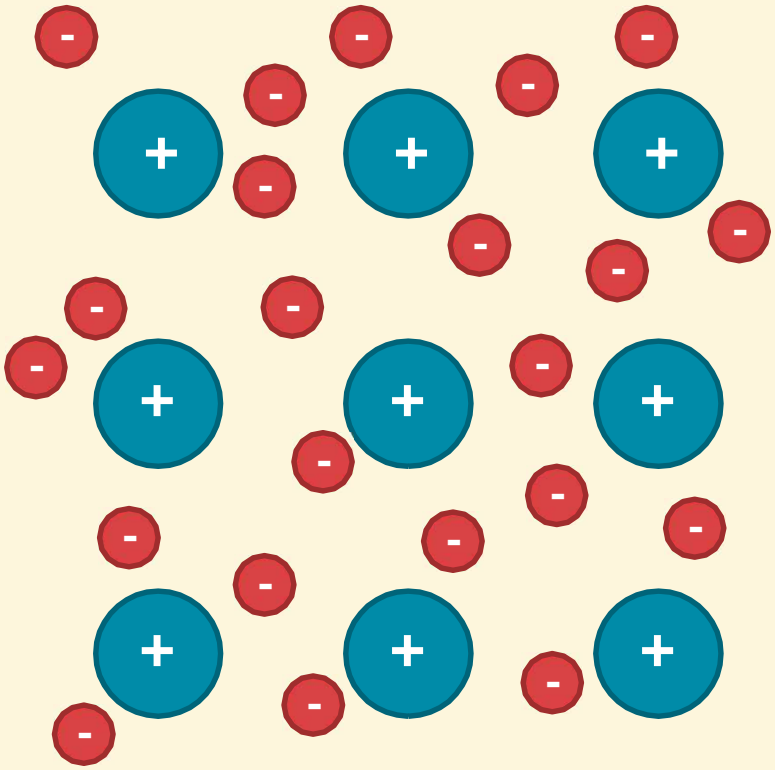



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# Metallic bonding

- **Mobile electrons** around fixed metal cations:
  - e.g. Metals!
  - Mobile electrons give the highest electrical conductivity

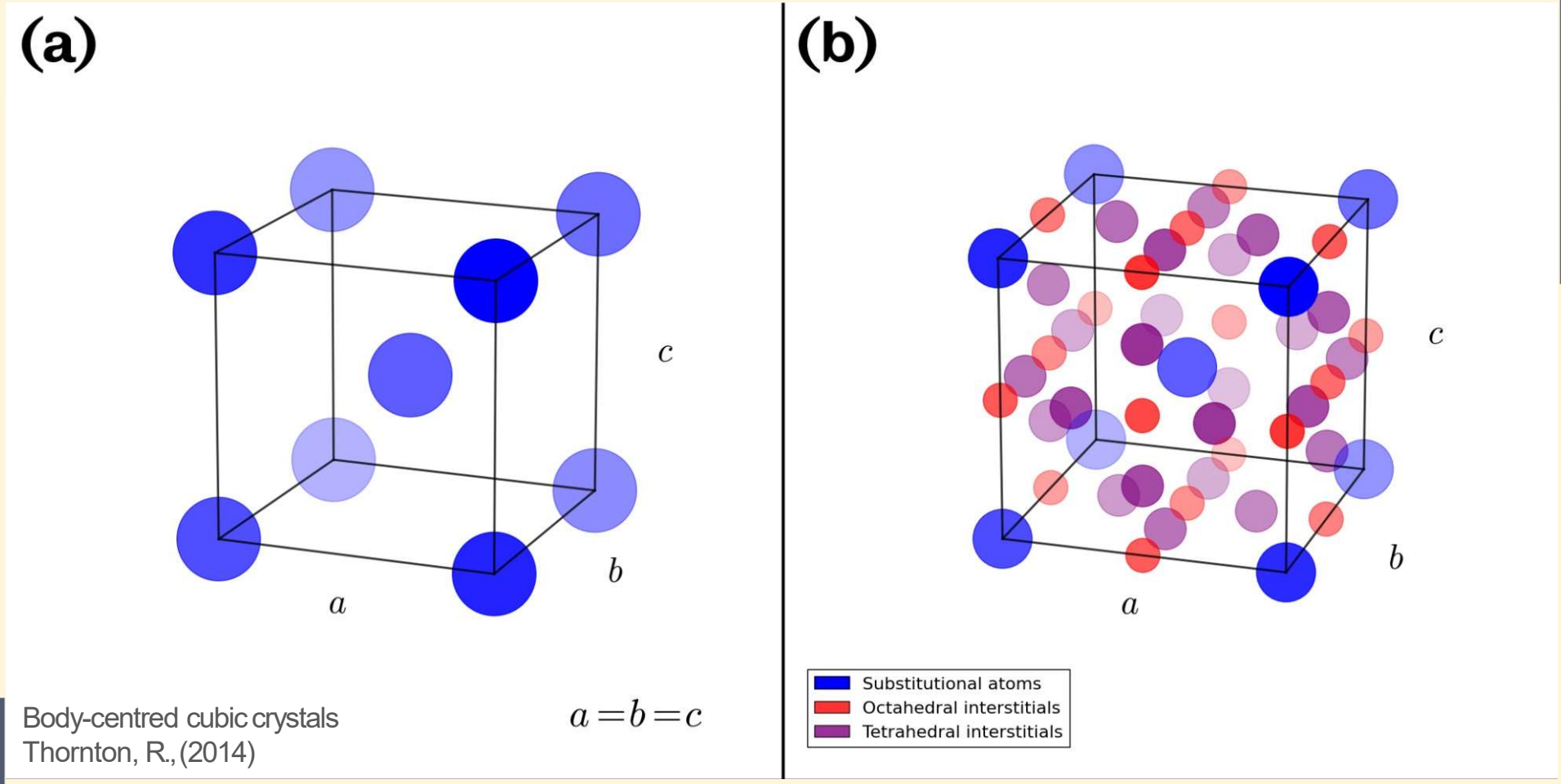
- Metallic bonds are completely **non-directional**:
  - Atoms can 'stack' close together while electrons flow around them



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# Crystal structures (more detail next time!)

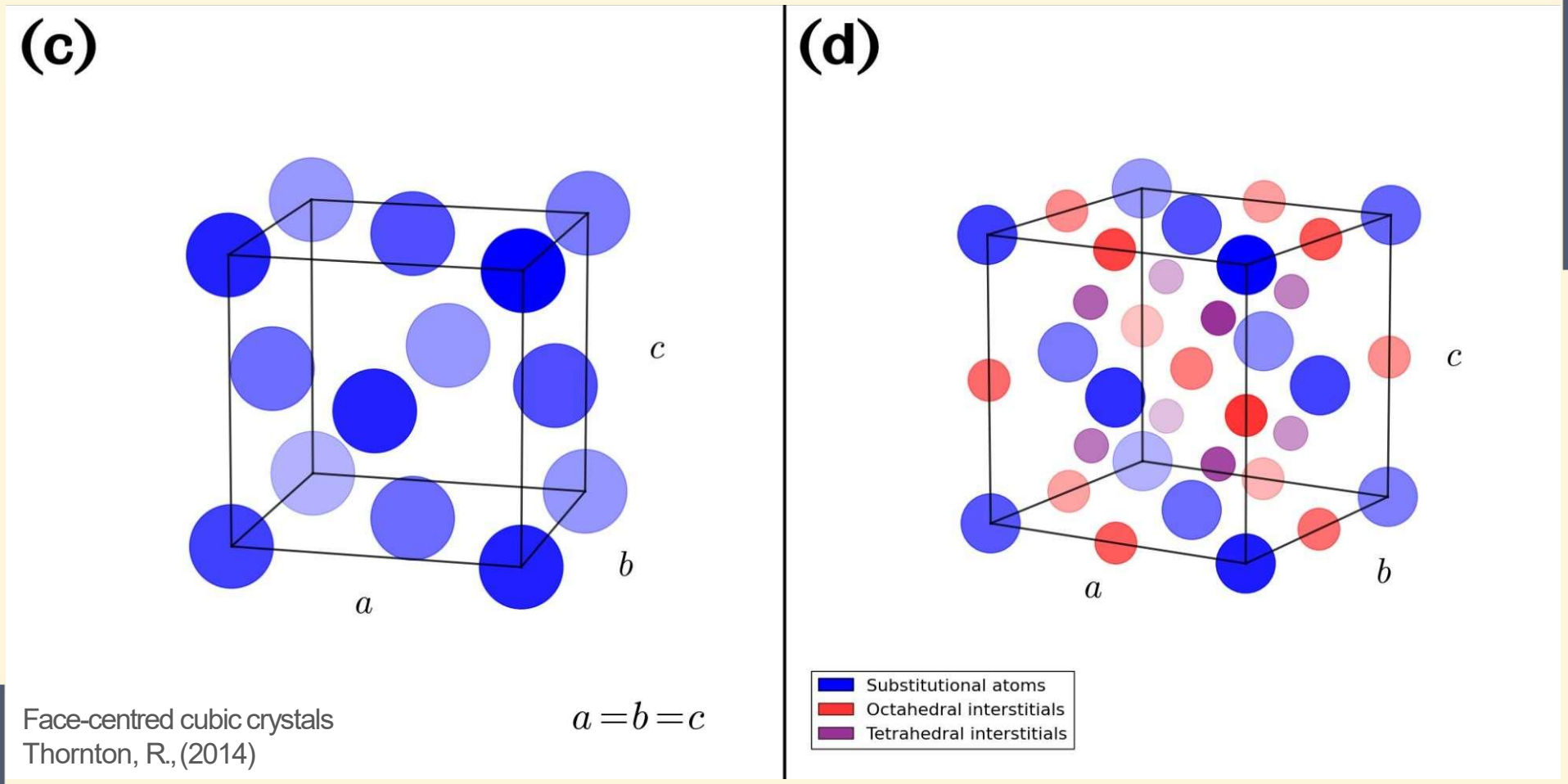
- Atoms in ceramics and metals organise themselves into **very ordered structures**, known as **crystals**
- Their arrangements and stacking have an enormous **impact on material properties** and behaviour
- Some common crystal structures and examples:
  - **Body-centred cubic** – e.g. Iron (ferrite), chromium, tungsten
  - **Face-centred cubic** – e.g. Iron (austenite), aluminium, copper, nickel
  - **Hexagonal close-packed** – e.g. Tungsten carbide (WC), zinc, titanium, magnesium
  - **Body-centred tetragonal** – e.g. Iron (martensite)
  - Other structures include **Orthorhombic**, **Rhombohedral**, **Monoclinic**, **Triclinic**…




Body-centred cubic crystals
Thornton, R., (2014)

$$a = b = c$$

# Crystalline structures – body-centred cubic




Face-centred cubic crystals
Thornton, R., (2014)

$$a = b = c$$

# Crystalline structures – face-centred cubic

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# Secondary bonding

- In Chemistry, also known as 'inter-molecular' bonding

- Much **weaker** than primary bonds…
  *…but still has a significant influence on properties*

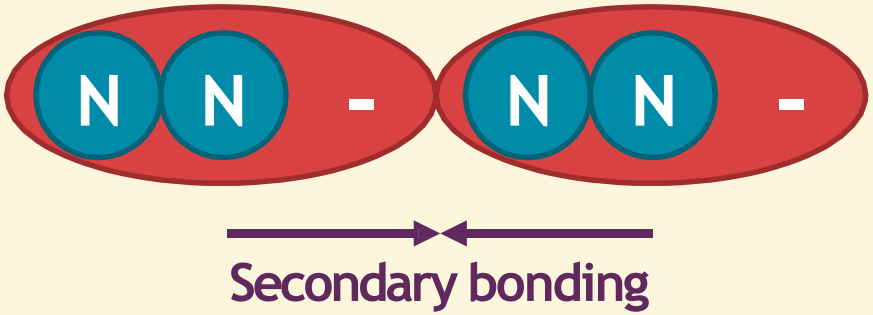
- Secondary bond types:
  - **Van der Waals** forces – dipolar attraction between uncharged atoms
  - **Hydrogen** bonding – dipole attraction between bonded hydrogen (slightly +ve) and neighbouring ions (slightly –ve); nitrogen, oxygen and fluorine are the only possible hydrogen bonding pairs

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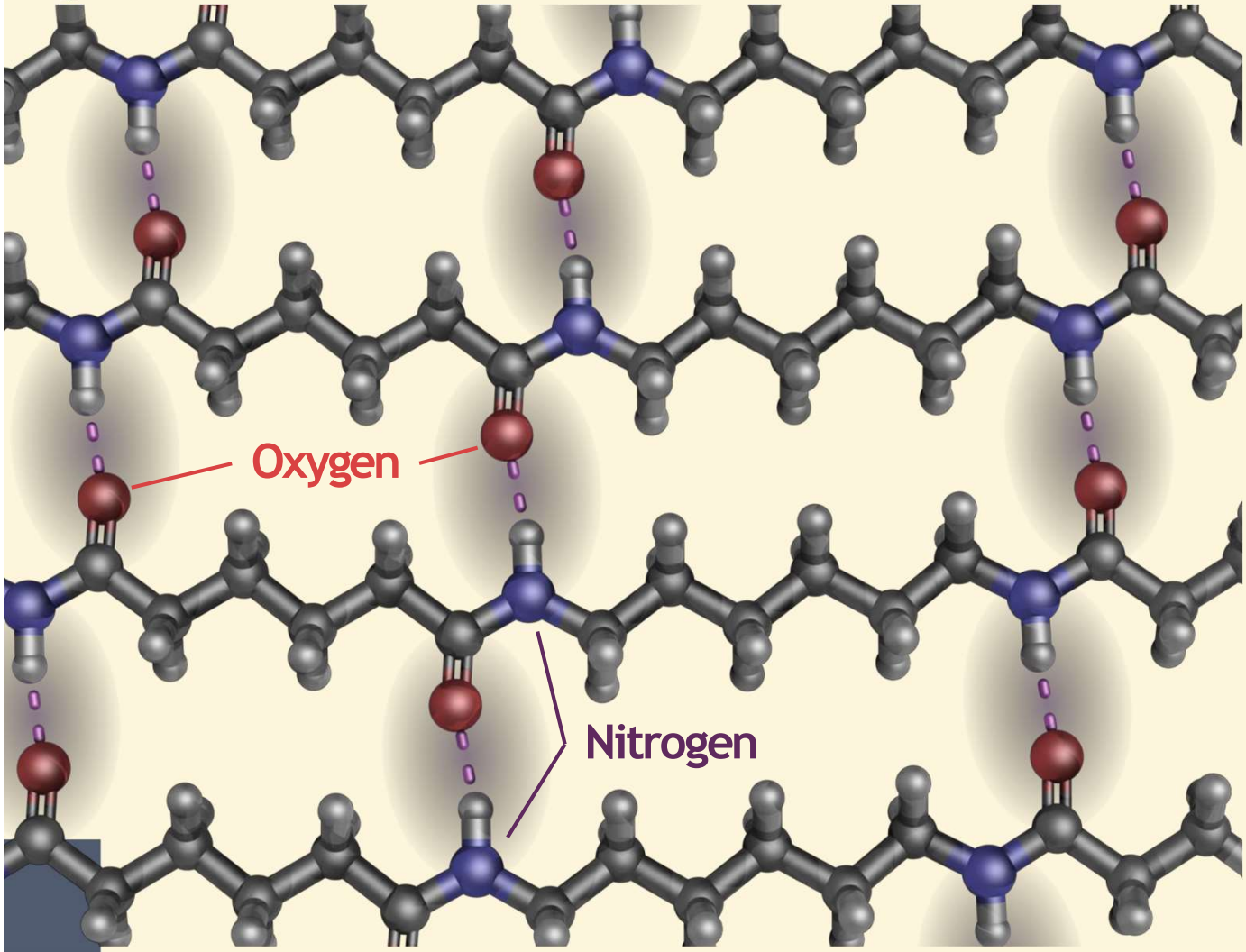
# Van der Waals forces

- **Dipolar attraction** between uncharged atoms
  - Electron movement causes **fluctuation in charge** on individual atoms, which then interact to create dipoles on others

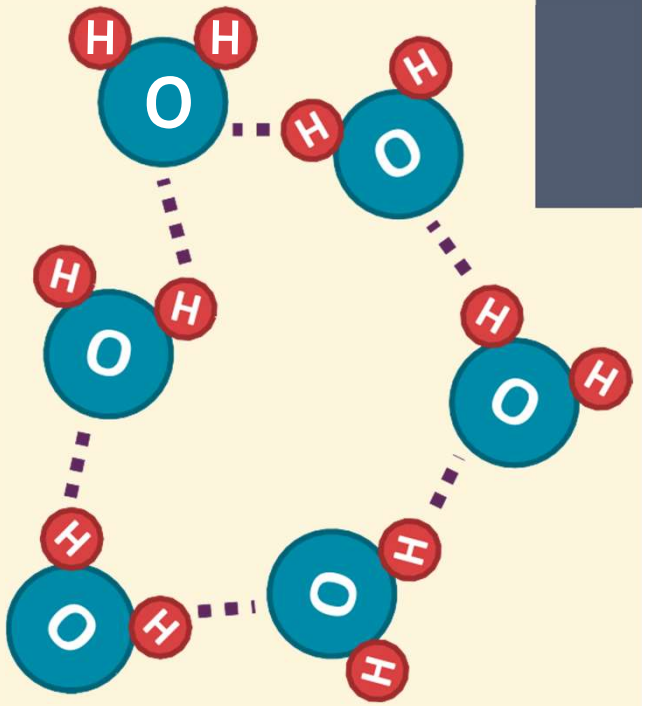
- Without Van der Waals forces, **liquefaction** would not be possible for many important industrial liquids:
  - Liquid nitrogen, oxygen, helium and hydrogen



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Hydrogen bonding in nylon6,6
https://commons.wikimedia.org/wiki/File:Nylon-3D-h_bond.png



# Hydrogen bonding

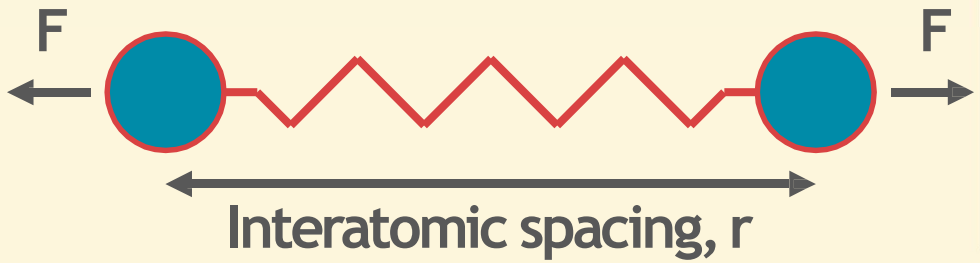
- Hydrogen atoms are able to **share their charge**:
  - Only with nitrogen, oxygen or fluorine in molecules
- Essential for solid polymers and keeping water liquid!

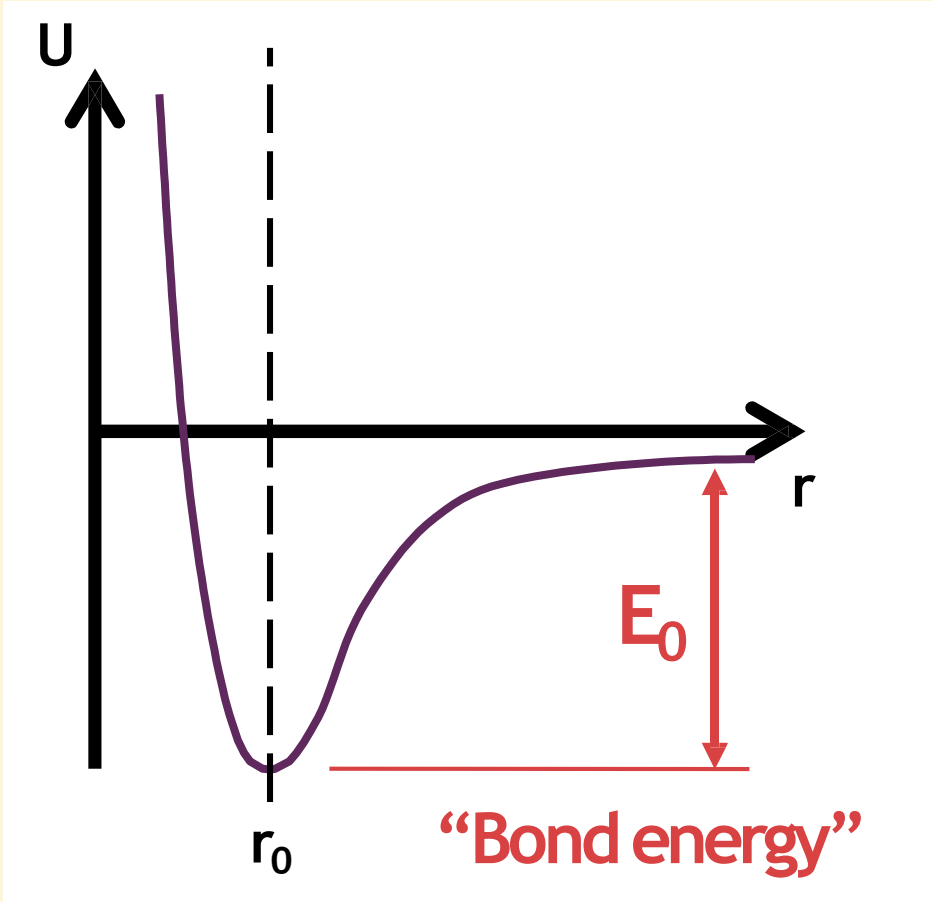
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# Bonding energies and properties

- Greater bond energy:
  - Greater stiffness (*E*)
  - Greater strength ($\sigma$)
  - Higher melting point ($T_m$)
    *Lower thermal expansion coeff. (a)*

- Generally speaking*:
  - Stiff, brittle materials are **ionically** bonded (e.g. ceramics)
  - Strong, ductile materials are **metallically** bonded (e.g. metals)
  - Good insulators are **covalently** bonded (e.g. polymers)




*Note that there are A LOT of exceptions to these generalisations!

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# Bonding types and properties

- We often talk of materials as though they only contain one type of bond
- In reality most materials have **several different bond types** within them:
  - e.g. Water ($H_2O$) molecules are covalently bonded, but hydrogen bonds exist between molecules
  - Many metals will feature covalent bonds as well as metallic
- Material **structure** also plays a significant role:
  - e.g. Diamond and graphite are purely made of covalently bonded carbon atoms, but their properties are *a bit* different!

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# Bonding types and properties

- **Ceramics**:
  - Large bond energy
  - Stiff and strong, but brittle
  - High melting point and low thermal expansion coefficient
  - **Ionic** and **covalent** bonding
- **Metals**:
  - Variable bond energy
  - Moderate stiffness and strength, can be brittle or ductile
  - Tend to have high thermal and electrical conductivity
  - **Metallic** bonding
- **Polymers**:
  - Directional properties
  - Low stiffness and strength; poor thermal properties
  - **Covalent** and **secondary** bonding

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# Concepts and terminology

- Atoms, ions, electrons
- Primary bonding:
  - Ionic
  - Covalent
  - Metallic
- Secondary bonding:
  - Van der Waals forces
  - Hydrogen bonding
- Bond energy
- Dipolar attraction
- Liquefaction
- Crystal structures:
  - Body-centred cubic
  - Face-centred cubic